RESEARCH BRIEFING

Projecting the OBIS eDNA DwC-DP example into DwC-A via the GBIF pipeline

How (and whether) occurrences are inferred
from nucleotide analysis rows

Source package: `gbif/dwc-dp-examples · edna/obis-edna-expeditions`
Mapping code: `gbif/pipelines · …/dwcdp/mapping/config` (`dev`)

Prepared 20 October 2026

1 Executive summary

Bottom line

Conceptually, the Darwin Core Conceptual Model treats Material + Nucleotide Analysis + Sequence (+ Identification) as **evidence from which an Occurrence may be inferred**. **In code today**, GBIF's DwC-DP → DwC-A recipes **do not** perform that inference: Occurrence rows are emitted only from an `occurrence` table. The OBIS eDNA Expeditions example omits that table (detections live as analyses). Under current `EventDwcaMapping`, the package would index **sampling events** (plus event-level assertions), not densified ASV occurrences — and material-linked analyses also miss the Event-scoped DNA fragments. Public 2026 plans (Work Programme, eDNA roadmap workshop) aim at DwC-DP as the DNA publish model and at **sequence-based occurrence search**, but they have not yet shipped a pipelines fragment that mints Occurrences from analyses.

The input DwC-A was Occurrence core + DNADerivedData. DwC-DP remapping is richer for publishing; the **inference** step the Conceptual Model describes is still a product/engineering gap between "store the graph" and "show detections in occurrence search."

2 The example package

The OBIS eDNA Expeditions example maps two Calanques field samples (`OBIS_EDNA_140`, `OBIS_EDNA_142`) from an Occurrence-core + DNADerivedData export into DwC-DP tables shaped like

the GBIF Metabarcoding Data Toolkit (MDT) export, **plus** a `material` stub required by the examples repo SQL schema.

Table	Role in this package
`event`	One row per field sample (`sample_id`).
`event_assertion`	Sample size, replicate, `env_medium`.
`material`	One environmental-DNA sample per event (stub; MDT omits this).
`molecular_protocol`	One row per distinct assay / primer set.
`nucleotide_sequence`	One row per distinct ASV sequence.
`nucleotide_analysis`	One row per **sample** × **ASV** detection (`read_count`). Links event, material, sequence, protocol.
`identification`	Taxonomy; linked to `nucleotide_sequence` only (not to analysis or occurrence).

Table 1: Tables present in the example (no `occurrence`).

`identification` —based on→ `nucleotide_sequence`
`material` —collected during→ `event`
`event_assertion` —about→ `event`
`nucleotide_analysis` —of→ `material`
`nucleotide_analysis` —material collected during→ `event`
`nucleotide_analysis` —produced→ `nucleotide_sequence`
`nucleotide_analysis` —followed→ `molecular_protocol`

Figure 1: Publisher graph for the example. Dashed material edges are the stub added for the SQL schema.

Mapping detail that matters for projection (from `map_obis_edna_input.sql`):

- `nucleotide_analysis.nucleotide_analysis_id` = original `occurrenceID` (so the **semantic** detection ID from DwC-A is preserved on the analysis row, not on an occurrence).
- `nucleotide_analysis.material_entity_id` = sample id (always set).
- `identification.nucleotide_sequence_id` only — no `occurrence_fk`, no `nucleotide_analysis_fk`.

3 How GBIF projects DwC-DP to DwC-A

GBIF does not index the DwC-DP graph as-is. `DwcDpVerbatimConverter` picks a **governing plan**, compiles/prunes fragments against `datapackage.json`, runs Spark joins, and writes DwC-A-shaped `verbatim.avro` (`ExtendedRecord`) for the existing interpretation stack. Background: project briefing `reports/gbif-cross-dataset-dwcdp.typ`.

Condition	Plan
Event-bearing crawl **and** package has `event`	`EventDwcaMapping.current` — Event core; occurrences only as **Occurrence extension** under events
Else if package has `occurrence`	`OccurrenceDwcaMapping.current` — Occurrence core
Otherwise	Empty verbatim + warning — no "index whatever exists" fallback

Table 2: Governing-plan choice (`DwcDpVerbatimConverter`).

This example has `event` and **no** `occurrence` → **Event-core plan**. Nested Occurrence rows come exclusively from fragments that start at the `occurrence` table.

4 Are occurrences inferred from nucleotide analyses?

4.1 Short answer

No. There is no fragment that mints an Occurrence from `nucleotide_analysis`, `nucleotide_sequence`, or `identification`.

4.2 Where Occurrence rows come from

Under `EventDwcaMapping.current`, the Occurrence extension is opened and seeded by:

> **Occurrence seed fragment**
>
> `OccurrenceMapping.directOccurrence` — path root = `occurrence`; `scopeKey = event_fk`; `rowIdentity = occurrence_pk`. Direct DwC fields are copied; `occurrenceID` falls back to `gbif:dwcdp:occurrence:occurrence_pk:…` when blank.

Every other Occurrence fragment (`acceptedIdentification`, `material`, `eventIdentity`, basisOfRecord fallbacks, …) **enriches** or **matches** an existing `occurrence_pk`. They do not create rows.

If the `occurrence` resource is absent from `datapackage.json`, the dataset pruner drops the whole Occurrence-extension branch. **Result for this example: zero Occurrence extension rows.**

4.3 What DNA fragments actually do

`NucleotideMapping` projects analyses into the **DNADerivedData** extension (`http://rs.gbif.org/terms/1.0/DNADerivedData`), not into Occurrence:

Fragment family	Scope	Path / filter
`eventAnalyses` (+ sequence / protocol)	`event_pk`	`event → nucleotide-analysis` via `event_fk`, **only if** `materialEntity_fk` **is null**; fan-out
`materialAnalysesForOccurrence` (+ …)	`occurrence_pk`	`occurrence → material` via `evidenceForOccurrenceID` (exactly one) → `nucleotide-analysis` via `materialEntity_fk`

Table 3: DNA fragment families and their parents.

`EventDwcaMapping` imports **only** the event-owned DNA fragments. `OccurrenceDwcaMapping` imports **only** the material-for-occurrence DNA fragments. There is no third path "analysis with material but no occurrence → invent an occurrence."

4.4 Identification fragments for DNA

`IdentificationMapping` similarly distinguishes:

- **Event-owned** DNA IDs: analyses with `materialEntity_fk` null, then identification via `nucleotideAnalysis_fk` or (weaker) via sequence with `nucleotideAnalysis_fk` and `occurrence_fk` null.
- **Occurrence / material** DNA IDs: require an occurrence with exactly one evidence material, then analyses on that material.

Again: enrichment of Identification **extension** rows under Event or Occurrence — not synthesis of Occurrence cores.

5 Walk-through: this package under `EventDwcaMapping`

Target	Emitted?	Why
Event core (2 samples)	yes	`event` table present; direct event fields + parent/protocol enrichments as available.
Occurrence extension	**no**	No `occurrence` table → `directOccurrence` pruned.
DNADerivedData on Event	**no**	`eventAnalyses` requires `materialEntity_fk` **null**; example always sets material.
DNADerivedData on Occurrence	**no**	Needs occurrence parent; Event plan does not import material-for-occurrence DNA fragments.
Identification (DNA paths)	**no**	Event DNA ID paths also require material-null analyses; sequence-linked IDs in the example never attach via those filters.
eMoF from `event_assertion`	yes	`AssertionMapping.eventAssertions` — sample size / replicate / env_medium.
Material as searchable core	**no**	Material never becomes its own GBIF.org index type; only enriches occurrences or feeds extensions.

Table 4: What survives pruning for `obis-edna-expeditions` as published.

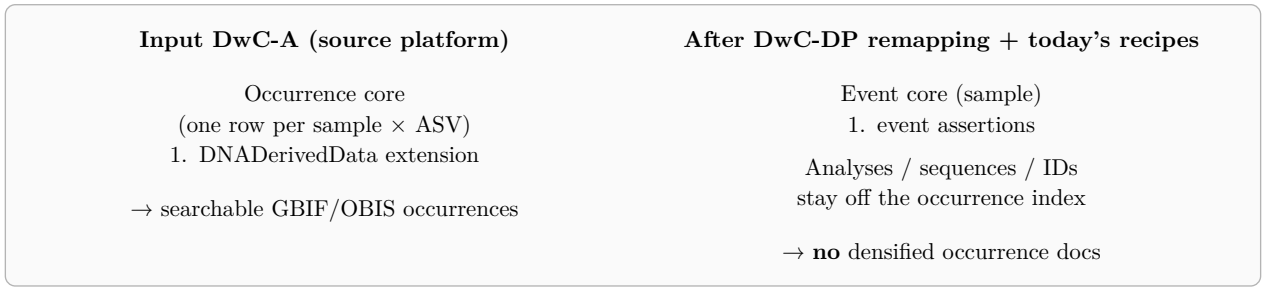


Figure 2: Intended DwC-A densify path (publisher mental model) vs current pipeline behaviour.

So the critical inference question — "does each `nucleotide_analysis` become an Occurrence?" — is answered by the **current** code: **not with the fragments that ship today**. The next section separates that fact from what standards and GBIF planning documents say **should** or **will** happen.

6 Plans for inferring Occurrences from DwC-DP

Public sources agree that DNA-derived **detections are meant to surface as Occurrences** for discovery, but they split across three layers: conceptual model, product roadmap, and pipelines engineering. They do **not** yet document a finished "analysis → Occurrence" recipe in `gbif/pipelines`.

6.1 Darwin Core Conceptual Model (what inference means)

The DwC Conceptual Model §2.6 (NucleotideAnalysis, NucleotideSequence, MolecularProtocol) is explicit:

- A Nucleotide Analysis of a Material Entity follows a Molecular Protocol; it may produce Sequences and/or confirm presence/absence of a target.
- Material, Analysis, and Sequence provide evidence for Identifications (morphological, assay-based, or reference-catalogue matches).
- **"The resulting body of evidence may be used to infer the Occurrence of an Organism in the sampled Material Entity"** — for qPCR-style detections and for barcoding / metabarcoding / metagenomics.

- **Inferred Occurrences should carry quality or confidence measures**; the CM does not prescribe how.
- If Material is not preserved or documented, implementers **may link the Nucleotide Analysis directly to the sampling Event** so spatio-temporal context (and thus Occurrence inference) remains possible.

DwC-DP is the reference tabular implementation of that model. So **omitting** an `occurrence` table in a metabarcoding package is a legitimate DwC-DP publishing choice **only if** some downstream process is allowed to **infer** Occurrences — which is exactly what today's Event/Occurrence mapping plans do not do.

Related CM language elsewhere: an Identification based on a Nucleotide Analysis "may subsequently be used to infer an Occurrence"; Occurrence itself covers direct observation, **indirect inference** that an Organism existed, or absence of detection.

6.2 GBIF product / DNA roadmap (2026 direction)

Source	What it commits (or strongly signals)
GBIF Work Programme 2026	Lead **sequence-based occurrence search**; improve DNA-derived filtering/downloads; prototype continuous taxonomic reinterpretation via reference catalogues; **implement DwC-DP processing** and richer visualisations; engage DNA infrastructures / eDNA community.
eDNA Data Infrastructure workshop (Copenhagen, 11–13 Nov 2025)	Endorse treating biodiversity data as **event-based**, with eDNA as a special case; **ultimately use DwC-DP as the primary model** for complex DNA datasets including NucleotideAnalysis and related CM tables; add a **sequence service layer** (clean + hash + non-redundant store); ASV→taxon index with publisher ID preserved + GBIF-mediated reinterpretations; DNA reference-library catalogue; mOTU prototyping; MDT/IPT/API publishing and ENA brokerage. Priorities include MIxS–DwC–DwC-DP alignment and DwC-DP review feedback on NucleotideAnalysis (OBIS contributing).
pipelines #1547 medallion store	Design Bronze/Silver/Gold lake; **Gold** example use cases include **DNA** or a **material catalogue** — i.e. specialised services over a multi-table store, not only flat occurrence docs.
pipelines #1543	Document how DwC-DP ingestion plugs into **existing occurrence–event pipelines** — confirms near-term path is still projection into that world.

Table 5: Public planning signals relevant to DNA → occurrence discovery.

Reading those together:

1. **Publish** complex DNA in DwC-DP (analyses, sequences, protocols, material, events) — not densified DwC-A forever.
2. **Discover** sequence-based evidence through occurrence-oriented (and later sequence-oriented) search services, with continuous taxonomic reinterpretation.
3. **Near-term engineering** still funnels DwC-DP through the occurrence/event verbatim path; a dedicated DNA "Gold" layer is sketched, not shipped as the default search surface.

What the roadmap documents **do not** yet specify in public detail: the exact join cardinality for "one Occurrence per `nucleotide_analysis` vs per distinct sequence vs per accepted identification," how confidence is encoded on inferred Occurrences, or whether inference runs at ingest (new fragments) vs in a later DNA service that **exposes** occurrence-like documents without writing classic `occurrence` rows into verbatim.

6.3 Publishing practice in examples (inference deferred to the package)

DwC-DP community examples often **materialise** Occurrences rather than relying on network-side inference. The Insektmobilen DNA survey use case in the DwC-DP implementation feature reports

maps Event, Material, Nucleotide Analysis/Sequence, Molecular Protocol, Identification, **and** Occurrence — and notes missing `occurrenceID`s as a blocker for test IPT publication. That pattern matches counterfactual A below: publisher-supplied Occurrence stubs keep today's recipes working while the CM's inference story matures.

The OBIS eDNA Expeditions example is the sharper stress test: it follows the CM/MDT-shaped graph **without** Occurrence, so it depends on future inference (or a synthetic fragment) to re-enter occurrence search.

6.4 Engineering gap (honest summary)

Layer	Status
Conceptual (DwC-CM)	Inference from Material/Analysis/Sequence/Identification → Occurrence is **in scope** and recommended to carry confidence.
Product (WP 2026 + eDNA workshop)	DwC-DP for complex DNA; sequence-based **occurrence** search; reinterpretation services; DwC-DP ingest — direction clear, recipe underspecified.
Pipelines mapping (`dev`)	**No** fragment mints Occurrence from `nucleotide_analysis`. DNA attaches as DNADerivedData / Identification **extensions** under Event (material-null analyses) or under Occurrence (via evidence material).

Table 6: Three layers of "infer Occurrence from DNA".

Interim historical note: pipelines #885 explored indexing MaterialEntity-core DwC-A **as if** Occurrence (`basisOfRecord` MaterialSample) for workshops — a different shortcut (material ≈ occurrence), not CM-style inference from nucleotide detections. DNA `basisOfRecord` values were earlier mapped to MaterialSample (#538) for filterability while modelling continued.

7 Counterfactuals (what would project today)

Understanding the gaps helps choose a publishing or engineering fix.

7.1 A. Keep an `occurrence` table (classic DNA-derived path)

If the package also published one Occurrence per detection (as the source DwC-A did), with:

- `occurrence.event_fk` → sample event,
- exactly one `material.evidenceForOccurrenceID` → that occurrence,
- `nucleotide_analysis.materialEntity_fk` → that material,
- identification accepted on the occurrence **or** reachable via the material DNA ID fragments,

then `OccurrenceDwcaMapping` (or Event-core nested Occurrence + DNA-on-occurrence fragments) would emit the familiar Occurrence + DNADerivedData shape. Taxonomy enrichment still prefers an accepted identification **on the occurrence**; sequence-only IDs need the DNA identification fragments to fire.

Cost: reintroduces densification / duplication that DwC-DP was meant to avoid at publish time — unless occurrences are thin stubs and heavy DNA stays in extensions.

7.2 B. MDT-style: analyses owned by Event, no material

If `materialEntity_fk` were left null and analyses hung only off `event_fk` (MDT omits material), then under Event-core:

- `NucleotideMapping.eventAnalyses` **would** fan DNADerivedData rows onto the Event,
- `eventDnaSequenceIdentifications` **could** attach Identification extensions to the Event when IDs hang off the sequence with the documented null filters,

…but those are still **Event extensions**, not Occurrence records. Occurrence **search** on GBIF.org would remain empty unless an Occurrence extension/core exists.

7.3 C. Implement the CM inference as a mapping fragment (aligned with plans)

A recipe that matches DwC-CM §2.6 and the Work Programme's "sequence-based occurrence search" framing could, for Event-core packages:

1. Path: `event → nucleotide-analysis` (fan-out; optionally via material — **including** non-null `materialEntity_fk`, unlike today's event DNA path),
2. Emit Occurrence extension/core row with `occurrenceID` = analysis id (or generated), `eventID` from parent, quantity from `read_count`, `basisOfRecord` suitable for DNA-derived detection,
3. Attach DNADerivedData from sequence + molecular protocol,
4. Attach / enrich Identification from sequence- or analysis-linked IDs,
5. Carry a confidence / QC field (CM requirement) — publisher score, reference-library match metrics, and later GBIF reinterpretation provenance.

That fragment **does not exist** in `NucleotideMapping` / `OccurrenceMapping` as of the `dev` tree reviewed here. Until it (or an equivalent DNA Gold service) ships, networks that need occurrence search from analysis-only packages must either add an `occurrence` table at publish time or run their own projector.

8 Implications for OBIS

For the OBIS indexing roadmap

1. **Today:** do not assume GBIF-like projection of the current eDNA DwC-DP example yields occurrence documents.
2. **Directionally:** DwC-CM + GBIF 2026 DNA plans **expect** inferred / sequence-based occurrences from DwC-DP molecular graphs — track fragment or Gold-layer design with GBIF rather than inventing a permanently divergent inference rule.
3. **Phase 2 choice:** Event+DNA search, densified Occurrence+DNA (publisher stubs **or** shared synthetic fragment), multi-table Parquet without densify, and/or a sequence service that **exposes** occurrence-like hits. Prefer aligning cardinality and confidence fields with CM §2.6.
4. The material stub is schema-faithful but **blocks** GBIF's only Event-scoped DNA path (`materialEntity_fk` must be null). If OBIS keeps material, the shared inference fragment must be the material-aware path (CM's preferred story), not the material-null workaround.

9 Sources

1. gbif/dwc-dp-examples — `edna/obis-edna-expeditions` (README, `map_obis_edna_input.sql`).
2. `EventDwcaMapping.java` · `OccurrenceDwcaMapping.java` · `OccurrenceMapping.java` · `NucleotideMapping.java` · `IdentificationMapping.java` (`dev` branch).
3. Darwin Core Conceptual Model §2.6 — inference of Occurrence from nucleotide evidence.
4. GBIF Work Programme 2026 — sequence-based occurrence search; DwC-DP processing.
5. GBIF eDNA Data Infrastructure: Technical Roadmap Review (Nov 2025 workshop).
6. pipelines #1543, #1547, #885, #538.
7. DwC-DP implementation feature reports — Insektmobilen / DNA tables including Occurrence.
8. Companion briefings in this repo: `reports/gbif-cross-dataset-dwcdp.typ`, project `README.md` (DwC-DP / eDNA section).

This briefing synthesizes public `gbif/pipelines` mapping configs, DwC-CM, GBIF Work Programme 2026, the Nov 2025 eDNA roadmap workshop report, and the published eDNA example (reviewed 20 October 2026). Internal GBIF designs not published online may differ. Typographic report generated with Typst.